



The 7th Annual

Parallel Applications Workshop Alternatives to MPI+X

Sunday, November 17, 2024
9:00 am - 5:30 pm (EST)

Hybrid Event

Held in conjunction with SC24
The International Conference for High Performance Computing, Networking, Storage, and Analysis





Parallel Applications Workshop
Alternatives to MPI+X

November 17th, 2024
Sunday



Panel Discussion: Alternative programming models for applications at scale

Panel Chair:

Christine Sweeney – Los Alamos National Laboratory

Panelists:

- Jan Ciesko – Sandia National Laboratories
- Nils Deppe – Cornell University
- Jason DeVinney – Center for Computing Sciences
- Eric Laurendeau – Polytechnique Montreal
- Julian Samaroo – Massachusetts Institute of Technology

Introductions



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Exceptional service in the national interest

Alternatives To MPI: Challenges and Lessons Learned

PAW-ATM, Supercomputing

Ciesko, et al.

November 17th, 2024

Sandia National Laboratories, New Mexico

SAND2024-15638C

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The Devil is in the Detail

- Experience obtained from two own ATMs and benchmarks
- Alternatives work great for convenience, but...

Stream benchmark (2 GPUs/node)

MPI 2-sided (CUDA-unaware)

```
struct mpi_add {
  ViewType_t A, B;

  mpi_add(ViewType_t A_, ViewType_t B_) : A(A_), B(B_) {}

  KOKKOS_FUNCTION
  void operator()(int i) const { A(i) += B(i); }

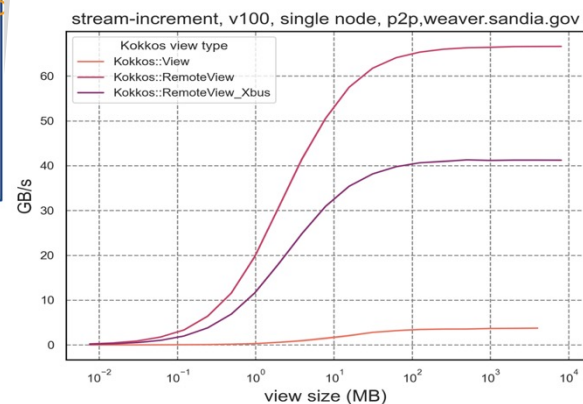
  void mpi_unaware_benchmark(int interval) {
    Kokkos::deep_copy(A, B);
    RemoteSpace_t().fence();
    if (comm.me == 0) {
      MPI_Isend(A.h.data(), interval, MPI_DOUBLE, 1, 1, comm.comm,
        &mpi_request_send);
      MPI_Wait(&mpi_request_send, MPI_STATUSES_IGNORE);
    }
    if (comm.me == 1) {
      MPI_Irecv(B.h.data(), interval, MPI_DOUBLE, 1, 1, comm.comm,
        &mpi_request_recv);
      MPI_Wait(&mpi_request_recv, MPI_STATUSES_IGNORE);
    }
    Kokkos::deep_copy(B, B);
    RemoteSpace_t().fence();
    if (comm.me == 0) {
      Kokkos::parallel_for("mpi_stream", policy_t(1), {interval}),
        mpi_add(A, B);
    }
    RemoteSpace_t().fence();
  }
};
```

Kokkos Remote Spaces

```
struct remote_add {
  RemoteView_t A;
  int my_min_i, rstart;
  remote_add(ViewType_t A_, int my_min_i, int rstart)
    : A(A_), my_min_i(my_min_i), rstart(rstart) {}

  KOKKOS_FUNCTION
  void operator()(int i) const { A(my_min_i + i) += A(rstart + i); }

  void remote_benchmark(int interval) {
    if (comm.me == 0) {
      Kokkos::parallel_for("remote_stream", policy_t(1), {interval}),
        remote_add(A, my_min_i, rstart);
    }
    RemoteSpace_t().fence();
  }
};
```



Halo exchange (Heat3D)

Heat3D with MPI

```
// Halo data
using buffer_t = Kokkos::View<double**,
  Kokkos::LayoutLeft, Kokkos::DefaultExecutionSpace>;
buffer_t T_left, T_left_out;

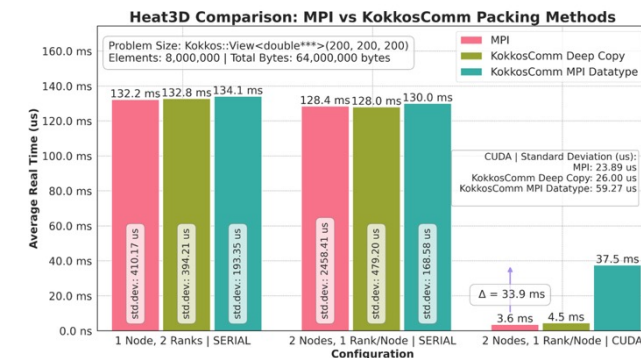
// ...
void setup_subdomain() {
  // incoming halos
  if (X_lo != 0) T_left =
    Kokkos::subview(T, T.extent(0) - 1,
      Kokkos::ALL, Kokkos::ALL);
  // ...
}

template <class ViewType>
void isend_irecv(int partner, ViewType send_buffer,
  ViewType recv_buffer,
  MPI_Request* request_send,
  MPI_Request* request_recv) {
  MPI_Isend(send_buffer.data(), send_buffer.size(),
    MPI_DOUBLE, partner, 1, comm, request_send);
  // ...
}
```

Heat3D with Kokkos Comm

```
// Halo data
using lr_buffer_t = Kokkos::Subview<DataView,
  int, decltype(Kokkos::ALL), decltype(Kokkos::ALL)>;
lr_buffer_t T_left, T_left_out;
// ...
void setup_subdomain() {
  // incoming halos
  if (X_lo != 0) T_left =
    Kokkos::subview(T, T.extent(0) - 1,
      Kokkos::ALL, Kokkos::ALL);
  // ...
}

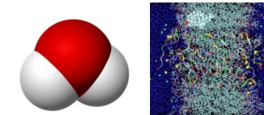
template <typename ExecSpace, class ViewType>
void isend_irecv(const ExecSpace &space,
  const ViewType &sv,
  ViewType &rv, int src, int dest,
  int tag,
  KokkosComm::Req &request_send,
  KokkosComm::Req &request_recv) {
  request_send =
    KokkosComm::isend(space, sv, dest, tag, comm);
  // ...
}
```



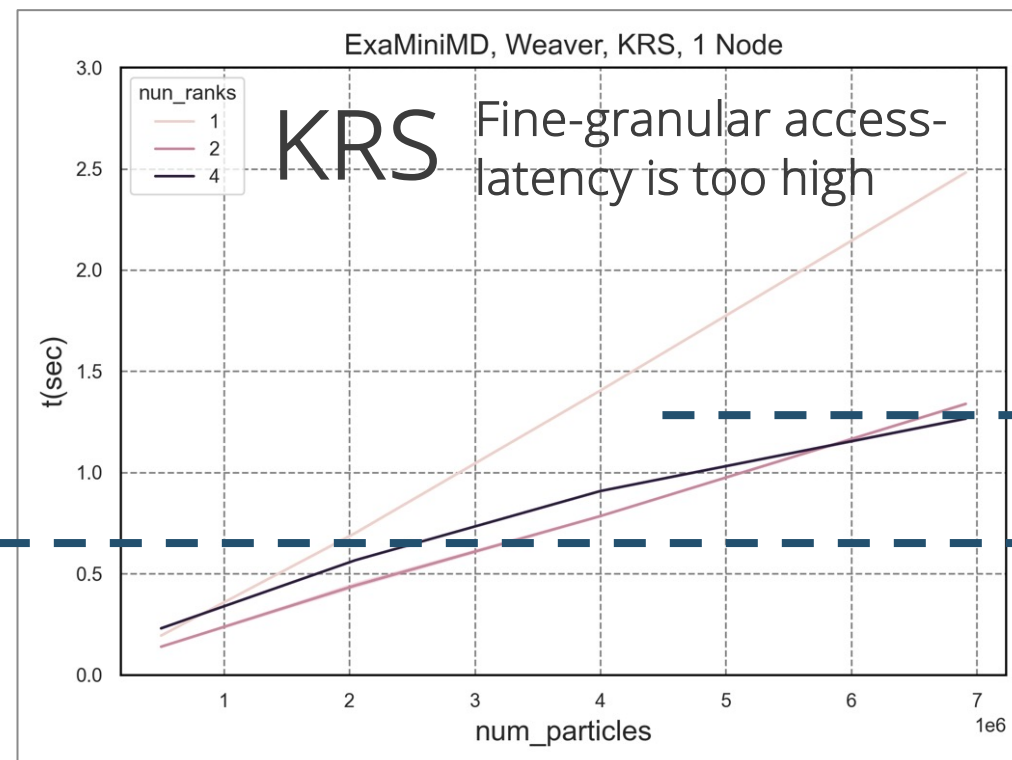
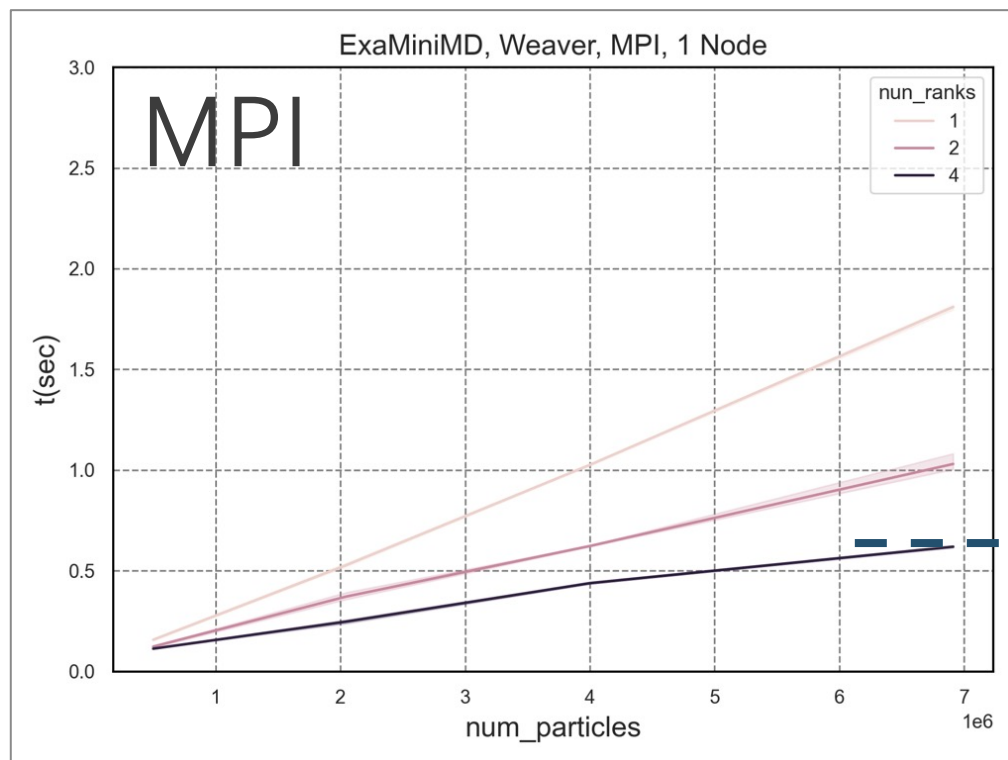


Solutions require high development effort

- Challenges: Fine-grained accesses vs. block transfers, API for data distribution, Vendor API issues, Lack of optimizations



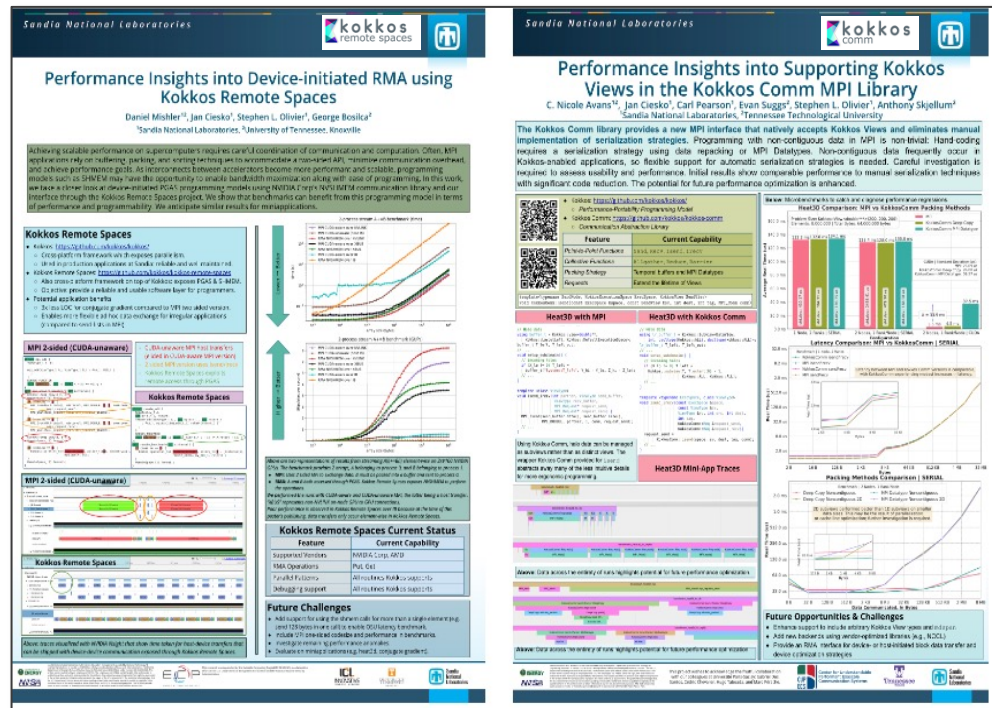
ExaMiniMD LJ Potential Kernel Only (ExaMiniMD, 1-4 V100 GPUs, Summit-like system)





The Challenge: Fragmentation, Quality and Flexibility

- Alternatives to MPI driven by MPI- and system software developers
- Alternatives to MPI driven by application and math library developers



<https://www.computer.org/csdl/proceedings-article/cluster-workshops/2024/834500a186/21EtNzVdjrW>
<https://www.computer.org/csdl/proceedings-article/cluster-workshops/2023/706200a066/1Shzgw7f2w>

Adoption likely requires a disruptive event in the HPC landscape that would require to rethink and reimplement communication.



Lessons Learned

- Co-design with application and library developers (We have enough to look at!)
- Quality of ATMs matters
- Disruptors and new trends: GPU-initiated communication, GPU-centric interconnects, fragmentation, productivity?
- We (CS folks) need to port the codes first
- Explicit and implicit Communication APIs
- Be vendor neutral and language embedded

Interested in Joining a Monthly call?

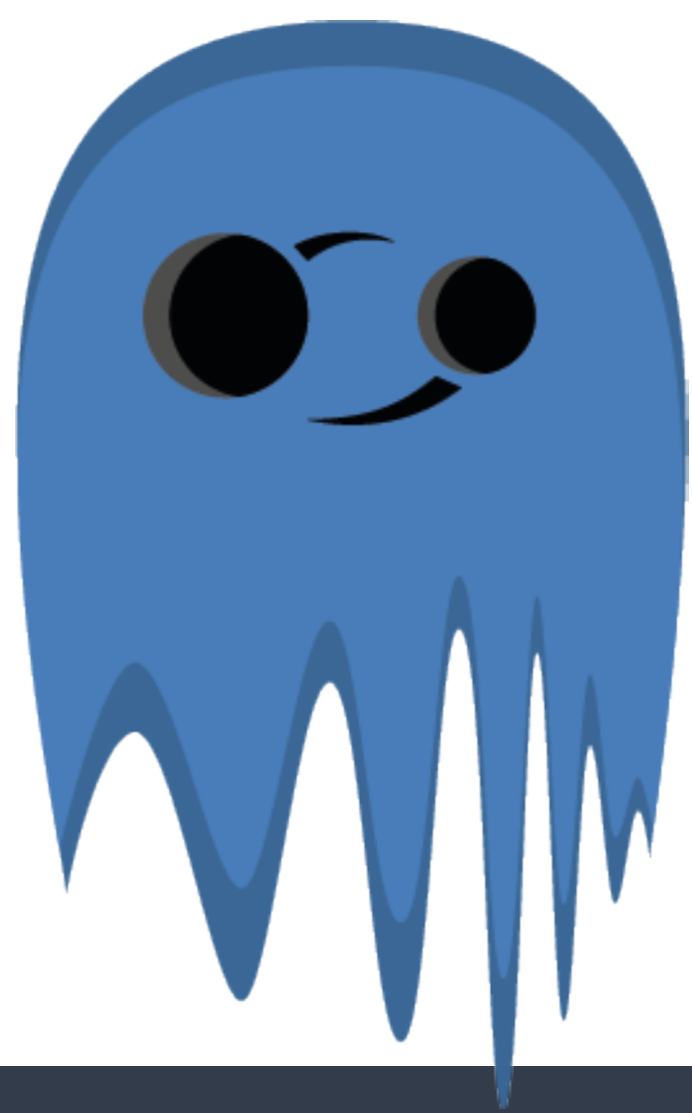
Application-driven Communication Abstractions for Exascale (ACAE)

Contacts:

jciesko@sandia.gov

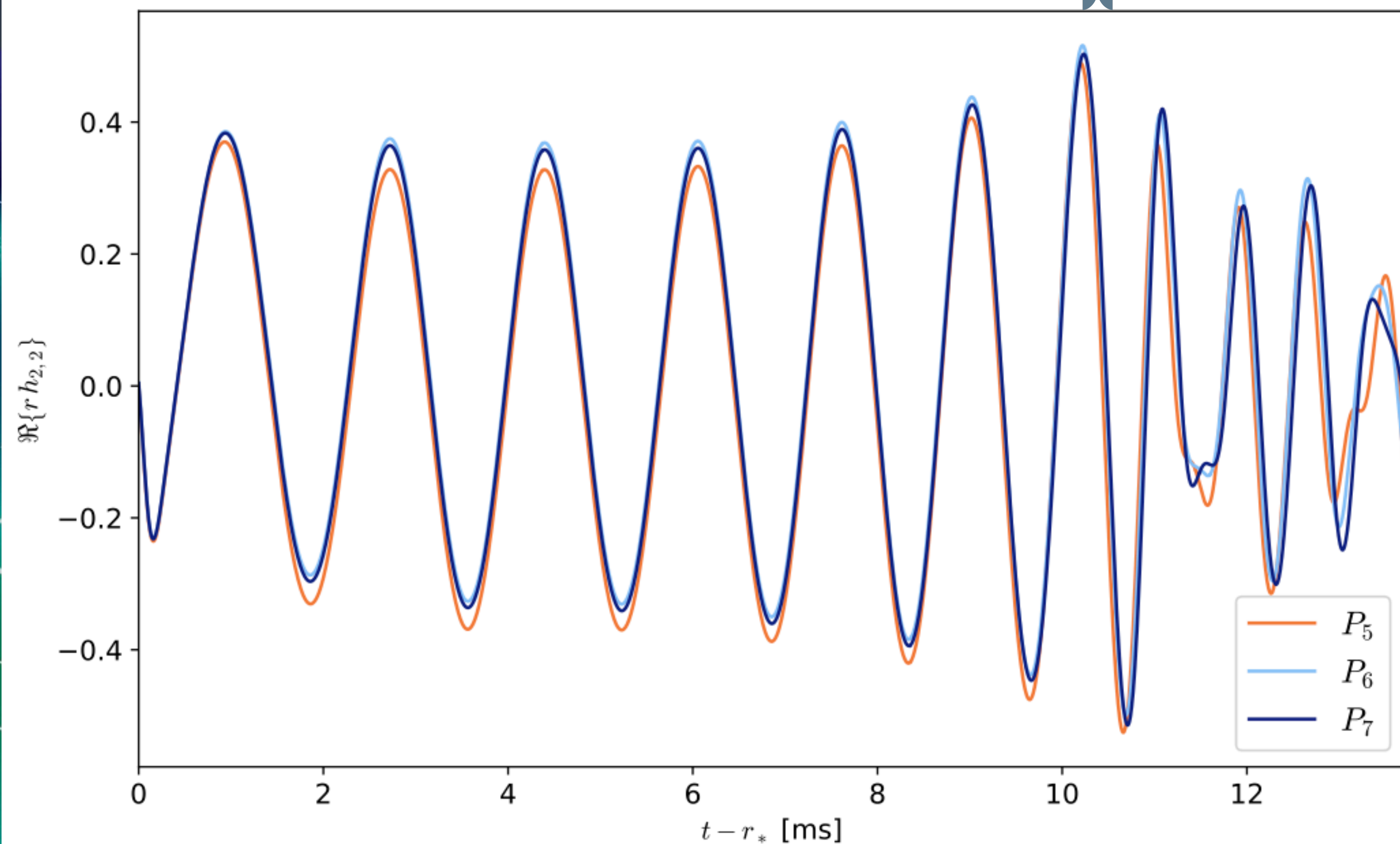
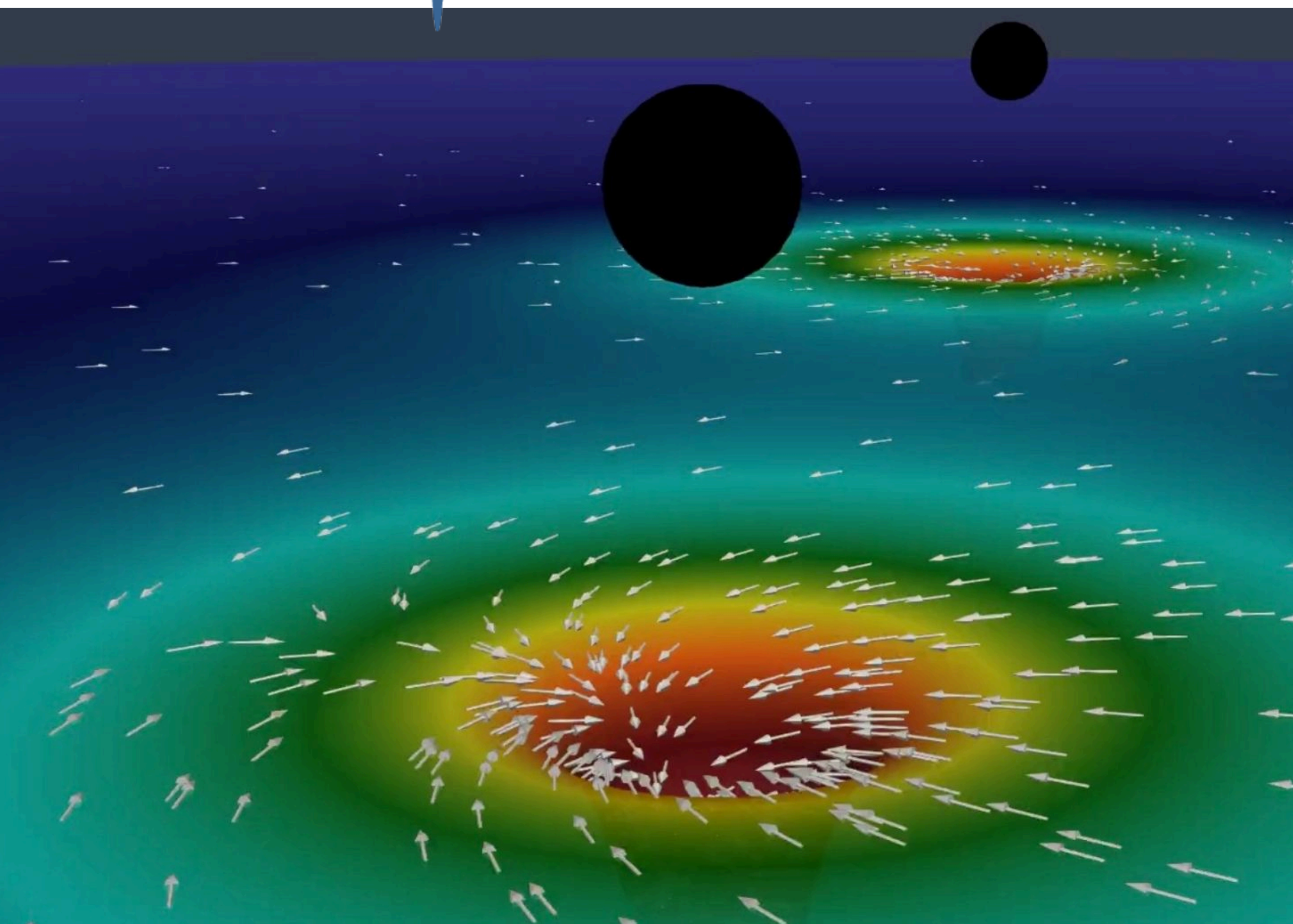
Partners:





spectre

Nils Deppe, Cornell University



arxiv: 2109.12033, 2109.11645, 2406.19038

github.com/sxs-collaboration/spectre

§pectre

Technologies:

- Spectral, finite difference, & Monte-Carlo
- Moving mesh
- Hyperbolic & elliptic PDEs
- Task-based/message-driven parallelism via Charm++
- Under development: GPUs via Kokkos

§pectre

Pros:

- Flexible physics
- Open-source
- Leverages other CI
- End-to-end pipeline
- Efficient algorithms

§pectre

Pros:

- Flexible physics
- Open-source
- Leverages other CI
- End-to-end pipeline
- Efficient algorithms

Cons:

- Technical guts
- Compile time
- Many features
- Technical algorithms

PAW-ATM Panel

SC 2024

Jason DeVinney – IDA/CCS
jgdevin@super.org

Hello!

I started at the Center for Computing Sciences (an FFRDC) in 2002.

I learned parallel programming, using UPC, on a Cray T3E.

I love developing new algorithms to solve problems and then implementing them, in parallel if necessary (it is usually necessary).

I migrated to SHMEM ~10 years ago.

I have been attending PAW since 2016!

I always feel like an outsider at SC... except at PAW-ATM.



Why ATM?

For me, I didn't really have much choice – the community had already chosen UPC before I got there.

No religious language war here. I just think parallel programmers should have it better.

Parallel programming is HARD and it is only getting harder as computing gets more complicated (CPUs/GPUs/smart NICS/modern interconnects).

There are not many people who are willing to brave all this and for this, our language development suffers.

github.com/jdevinney/bale

Pros and Cons of SHMEM

Pros		Cons
• simple one-sided memory model! • It's an open source library (no special compiler) • for many applications, PGAS languages are a more natural fit • extremely scalable and performant	• low level • C	• small user base • spec is getting complicated

Image Credits

- "Yes, it really IS a Cray T3E" by stiefkind is marked with CC0 1.0

MULTI-DISCIPLINARY AND MULTI-FIDELITY AERODYNAMIC MODELING FOR AIRCRAFT APPLICATIONS USING CHAPEL

ERIC LAURENDEAU
PAW-ATW 2024
NOV. 17, 2024

Mechanical Engineering Department

POLYTECHNIQUE
MONTRÉAL



BIOGRAPHIE

1996-2011

- Bombardier Aerospace, Advanced Aerodynamics Department
 - Numerical methods for analysis and design of aircraft, HPC (300th on top 500 list)
 - CRJ-700/900/1000, Challenger 300/605, Global Express/7000, C-Series (now Airbus 220)



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2011-now

- Professor, Polytechnique Montréal
 - Bombardier/CRIAQ/NSERC industrial Chair in Interdisciplinary Aerothermodynamic Analysis And Design Methods for Transport Aircraft
 - Canada Research Chair in interdisciplinary multifidelity aerodynamic aircraft analysis and design



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Paper 1



Paper 2



Paper n

Images: wikipedia



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Paper 1



Paper 2



Paper n

Images: wikipedia

+ students!



AERODYNAMIC SOFTWARE

CHAMPS-Chapel Multi-Physics Software

- Written by students
- Only known CFD software not using MPI/OpenMP in Aerospace Community
- Approx. 150 000 lines
- Competitive (but not better, even less accurate) to industrial software (Boeing, Airbus, Bombardier, etc.) and research centers (NASA, ONERA, DLR, JAXA, etc.)
 - More a demonstration software, not certified
- Can perform international workshops to baseline software performance
 - American Institute of Aeronautics and Astronautics DPW, HLPW, Icing, etc.

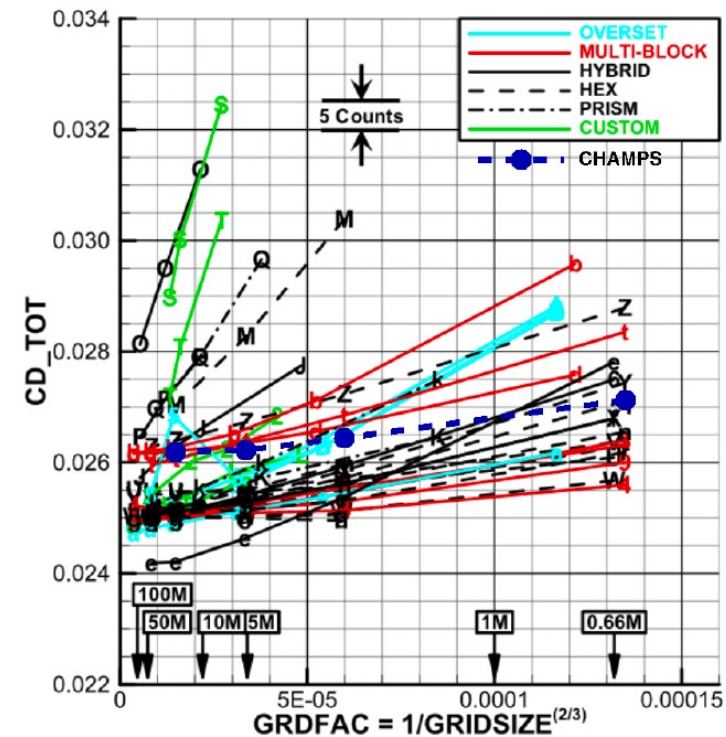
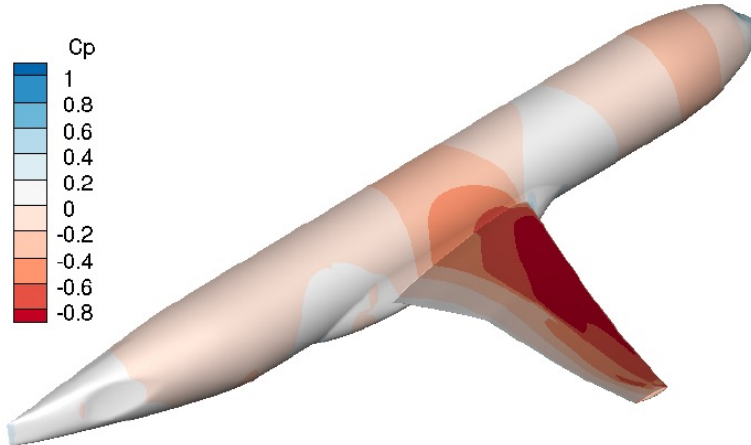


AERODYNAMIC SOFTWARE

DPW5

Drag Prediction Workshop 5

- Accurate prediction of total drag
- Results in line with other state-of-the-art flow solvers



AERODYNAMIC SOFTWARE

Pros:

- Perfect for laboratory with 'rotating' staff, writing parallel efficient HPC software
 - MSc 2 years, PhD 3-5 years.
- Parallel computing is simplified and more accessible with straightforward abstractions
- Advanced distributed memory management.
- Can very quickly write software libraries with HPC-type performance.
- Perspective for GPU compatibility

Cons:

- Not fully integrated with standard developer toolkits
 - Code profiler, debugger, etc.
- Compiling time is order of minutes
- Still small community of practice, not easy to ask around...counterbalanced by exceptional HPe support from Dr. Chamberlain's team.



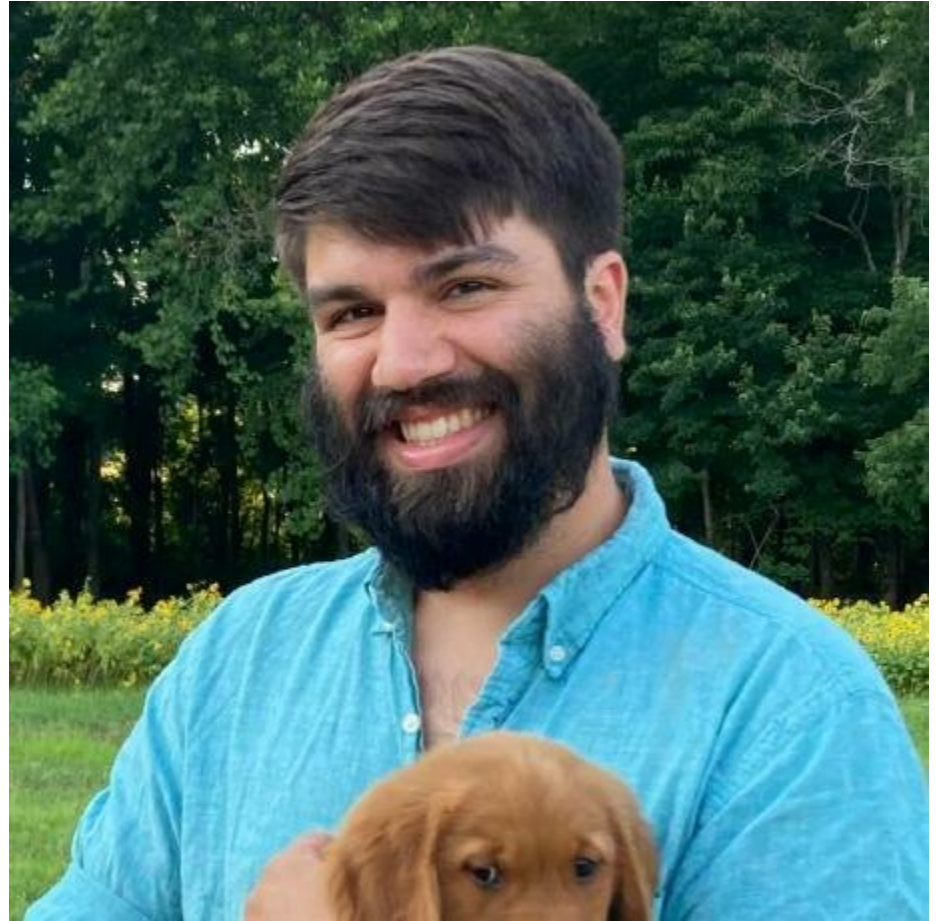
PAW-ATM Panel

Julian Samaroo (MIT JuliaLab)

Who am I?

I am a Research Software Engineer at MIT's JuliaLab. I work 95% on the Julia language, with a focus on developing libraries for distributed, multi-threaded, and GPU paradigms. I built and maintain Julia's AMDGPU support, and currently focus on developing the Dagger.jl HPC library.

And yes, that is my Golden Retriever in the picture :)



My MPI+X Alternative:



- Pure-Julia task parallelism library
- Unified task API supports multiple high-level APIs (arrays, tables, graphs, streaming, data dependencies)
- Supports multi-threading, multi-node, multi-GPU out of the box
- Machine learning-based scheduler optimizes computational DAGs based on learned task behavior
- Advanced data dependency system automatically parallelizes operations on overlapping data regions (like OpenMP's "task depend" on steroids)

Applications of Dagger.jl

- Quantum Tensor Network Contraction (BSC)
- HEP Streaming Framework (CERN)
- Reinsurance Pricing Engine (Ark)
- ...and more

Advantages

- Removes cognitive burden of manual task scheduling and data transfers
- No/minimal code changes to support different accelerators
- Scales automatically to available hardware
- Variety of built-in parallelized operations
- Multiple APIs for different problems

Known Issues

- Relatively high per-task overhead
- Missing optimizations for NUMA, optimal scheduling, many small tasks, etc.
- High memory allocations (count and size)
- Missing extensive (necessary) docs for new users
- Bus factor == 1 (approximately)



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Question 1:

How are ATMs exploiting changing supercomputer node architectures that are increasingly complex and heterogeneous, and may have many types of parallelism as well as memory spaces? (address recent hardware trends)

Audience Questions



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Question 2:

How are ATMs meeting the needs of, or in some cases adapting to, scalable applications in novel areas such as artificial intelligence, data analytics, and bioinformatics?

Do you foresee more or less collaboration with MPI+X for these applications? (address changing application landscape and some shifting to data-intensive application workloads)



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Question 3:

What have we learned about using ATMs in large (and even exascale) applications that can be applied going forward to meet the future requirements of scalable computing? (reflect on past experience)



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Question 4:

What is the role of ATMs for fast prototyping scalable applications?

In what scenarios are ATMs more appropriate than MPI+X for prototyping?

Some possible follow-ups: Do we really need ATM's just for prototyping — does one have to switch to MPI for "the real thing"? If so, what do ATM's need to do to overcome this?

And if they are only intended for prototyping, is there an easy path to moving from the prototype to MPI? (discuss adoption issues)



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Question 5:

Can AI/ML or advanced analytics help ATMs be more performant (eg. learn best data movement, computation placement, etc.), portable, and/or more widely used (e.g. LLMs for transpiling code from one programming model to another)?

If so, how?

If not, why not?

Thank you for attending!



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